

na jikar kar, na fikar kar
apne jazbaton ki, tu bas qadar kar
Jo tera hai, wo tujhe mil hi jaayega
aapne bari ka tu bas sabar kar

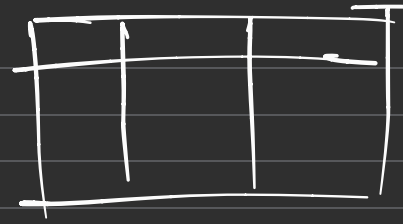
DATA BASE
SQL

SQL

- Structured Query Language
- Mainly used to interact with Database.
- Mainly Classified into three parts, i.e DDL, DML, DCL

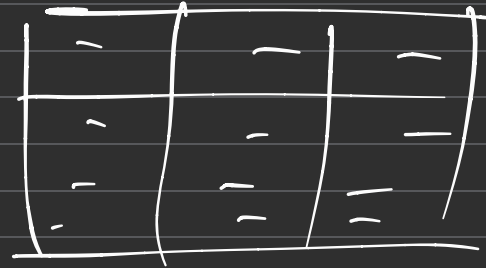
DDL (Data Definition Language)

- Mainly used to perform operation on Table.
- CREATE, TRUNACATE, DROP, ALTER



DML (Data Manipulation Language)

- Mainly used to perform operation on data of table.
- INSERT, DELETE, UPDATE, SELECT



-	-	-
-	-	-
-	-	-
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DCL (Data Control Language)

- Used to save Permanently.
- ROLLBACK, SAVEPOINT, COMMIT, GRANT, REVOKE

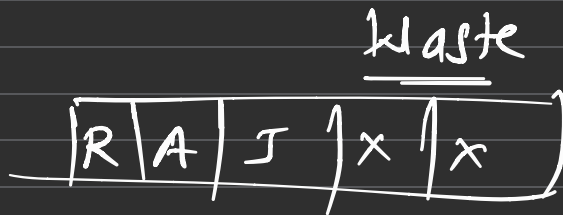
DATA TYPES IN SQL

- `number(size) / int (size)`

Mainly used to assign decimal values upto the size.

- `char(size)` $\bar{S}$

Mainly used to assign char values upto the size.



- `varchar(size)`

Mainly used to assign char values upto the size, but if string is less than the given size, it automatically get adjusted. It saves the space.



Data Definition Language

- **CREATE**

Mainly used to create the table.

* Syntax

```
CREATE TABLE <Name> (col, DT, col2 DT)
```

Eg:

```
CREATE TABLE Student (roll int, name varchar(20),  
marks int)
```

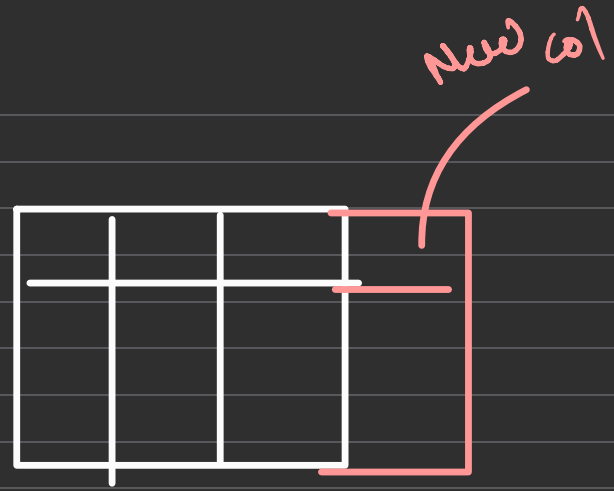
- ALTER

Mainly used when we want to add, delete any column or want to change the constraints.

Roll	Name	Marks	dept	Clg

Student

* How to add column



ALTER TABLE <Table Name> ADD <column Name> Data Type

* How to delete column

dept

please delete column

ALTER TABLE student DROP <column Name>

- DROP

Deletes the whole structure of table.

DROP TABLE <Table Name>

- TRUNCATE

Deletes all the rows. (Only delete the data)

TRUNCATE TABLE <Table Name>

Data Manipulation Language

- **INSERT**

Used to insert the data in the table.

int	varchar	int
Roll	Name	Marks

INSERT INTO <T.Name> VALUES (101, 'Yash', 70)

- UPDATE

Used to update the data or name of column in the table.

UPDATE student SET marks = 70

UPDATE student SET marks = 90
WHERE roll = 130

Roll	Name	mark
130	Sushant	90
—	—	—
150	Sushant	100

Student

update 70

- **DELETE**

Mainly delete all the content, but structure of the table remains, so in future we can still insert the data in the same table.

- **SELECT**

Mainly used to retrieve the data from the table.

Syntax:

SELECT * FROM TABLE_NAME

all columns

SELECT col1, col2 FROM Table_Name

Aggregate Functions (VVIMP)

All the aggregate functions are mainly applied on column. All the functions ignores the NULL value.

- **MIN()**

Returns the min value from the column.

- **MAX()**

Returns the max value from the column.

- **SUM()**

Returns the sum of all the values from the column. Must be on Integer column.

- **AVG()**

Returns the avg of all the values from the column. Must be on Integer column.

- **COUNT()**

Returns the number of rows.

	C ₁	C ₂	C ₃
	—	—	—
R ₁	—	Null	—
R ₂	—	—	—

SELECT MIN(name) FROM Student

SELECT MAX(marks) FROM Student

SELECT SUM(marks) FROM Student

SELECT AVG(marks) FROM Student

SELECT COUNT(roll) FROM Student

* MIN()

SELECT MIN(MARKS) from
student

* Sum

SELECT SUM(marks) from
student

Roll	Name	Marks
1	A	90
2	B	70
3	C	60
4	D	50

student

fun(x, arr)

LIKE

① % = any No. of characters

print all Names starting from x (x _)

SELECT Name from Student WHERE name like 'x%'

like {
x
x _
x _ _
x _ _ _
x _ _ _ _
x _ _ _ _ _

- Underscore _ = Single char

print name of student having second char
as x

[x] _ _

select name from student WHERE

name like '%x%' ✗

name like '_x%' ✓

Sub-Query

JOINS

Used to perform the cartesian product.

- CROSS JOINS
- NATURAL JOINS
- FULL OUTER JOINS
- THETA JOINS
- LEFT OUTER JOINS
- INNER JOINS
- RIGHT OUTER JOINS